Workforce Analysis Report using CRISP-DM Approach

Business Understanding

Overview

This report evaluates workforce data at a pan-African technology services company with offices across 4 countries in cities Nairobi, Mombasa, Kisumu, Kampala, Dar es Salaam, and Kigali. It has a workforce spread across six departments: Finance, HR, Operations, Sales, Engineering and Customer Support with 24 job roles spread across the departments.

Problem Statement

The business is looking to better understand its employee dynamics as they head towards tracking their employee headcount having noticed no clear records are tracked when it comes to employees despite having data from the year 2017 with the most recent entries being in 2025. The expanding workforce has made it difficult to monitor employee trends and they want to understand how best they can increase employee retention and overall performance as part of their workforce planning initiative. This analysis is the first step towards realising this goal.

Other Objectives

1. To know current employee headcount by office location, department, job role, education level, gender and age.
2. To analyse employee performance rating and training participation by department.
3. To know the average tenure of leavers and evaluate employee attrition patterns over time across gender, office locations, departments and education level.
4. To analyse salary distribution and diversity balance across the company's workforce.
5. To provide a detailed and summarised view of the company workforce.

Data Understanding

Data Source and Structure

The data source is an excel file which will eventually be imported to Power BI. I used the excel copy to fully understand the data before starting my analysis. The dataset contains 1000 rows of data with each row representing a unique employee and 16 columns. The 16 fields cover unique details about the employee.

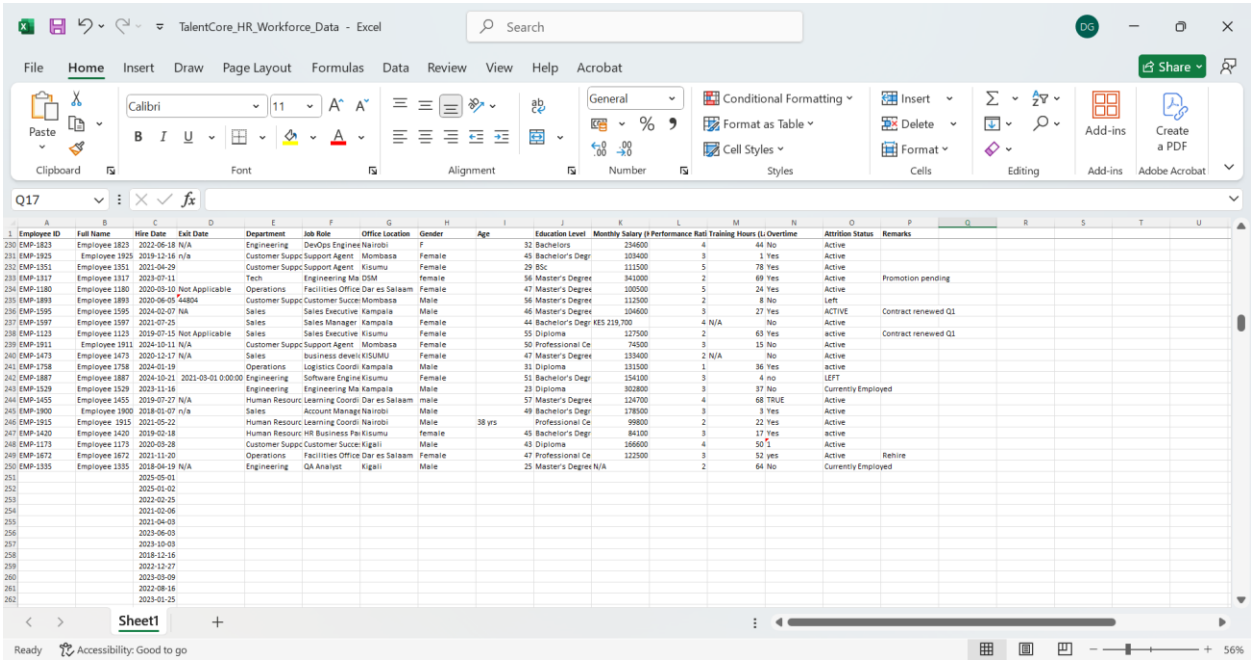
Key Fields

Field	Description
Employee ID	Unique identifier for each employee
Full Name	Name of the employee
Hire Date	Date the employee joined the company
Exit Date	Date the employee left the company (blank for active employees)
Department	Business unit (Finance, HR, Operations, Sales, Engineering and Customer Support)
Job Role	Specific role within the department
Office Location	City where the employee is based
Gender	Gender of the employee
Age	Age of the employee
Education Level	Highest qualification attained
Monthly Salary (KES)	Gross monthly salary
Performance Rating	Most recent rating on a 1 to 5 scale
Training Hours (Last 12 Months)	Hours of training completed
Overtime	Whether the employee regularly works overtime
Attrition Status	Whether the employee is Active or has Left
Remarks	Any notes on the employee

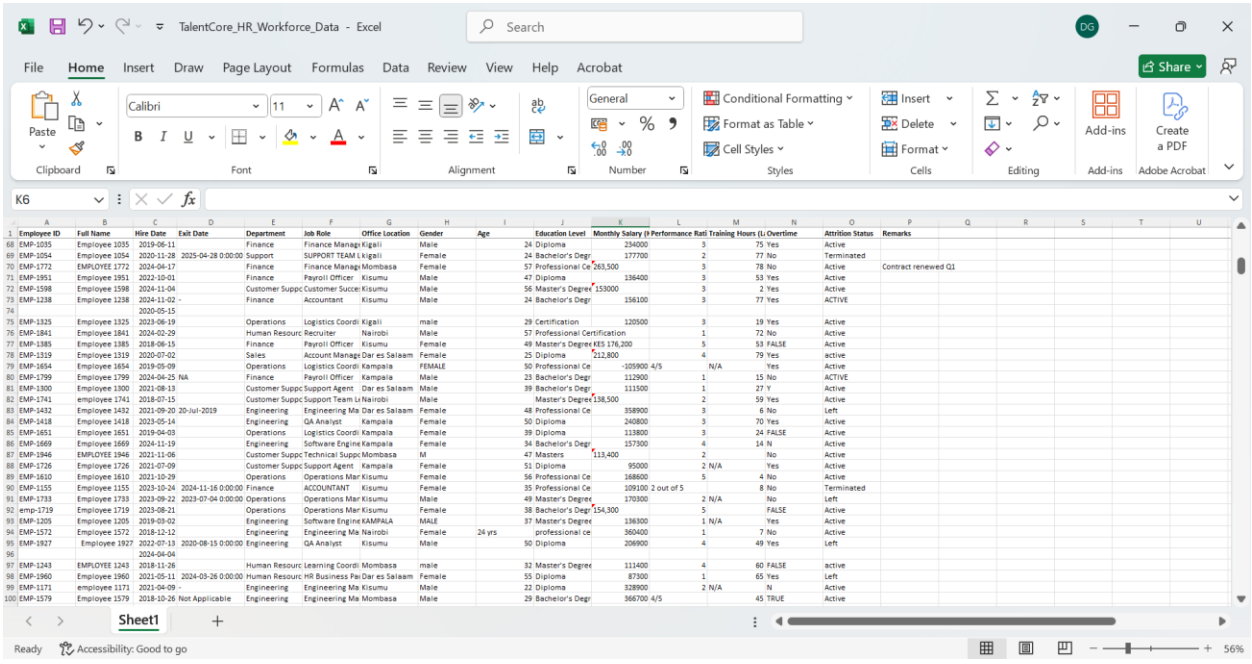
These fields are to help analyse the company's performance, employee characteristics and attrition patterns.

Data Quality Observations

1. With a total on 1000 rows, the dataset is largely incomplete with only 247 rows having most of data needed for an analysis. 751 rows only have the hire date field completed with all other fields empty.



2. 2 rows*(74 & 96)* was completely empty with just the hire date field tabulated.

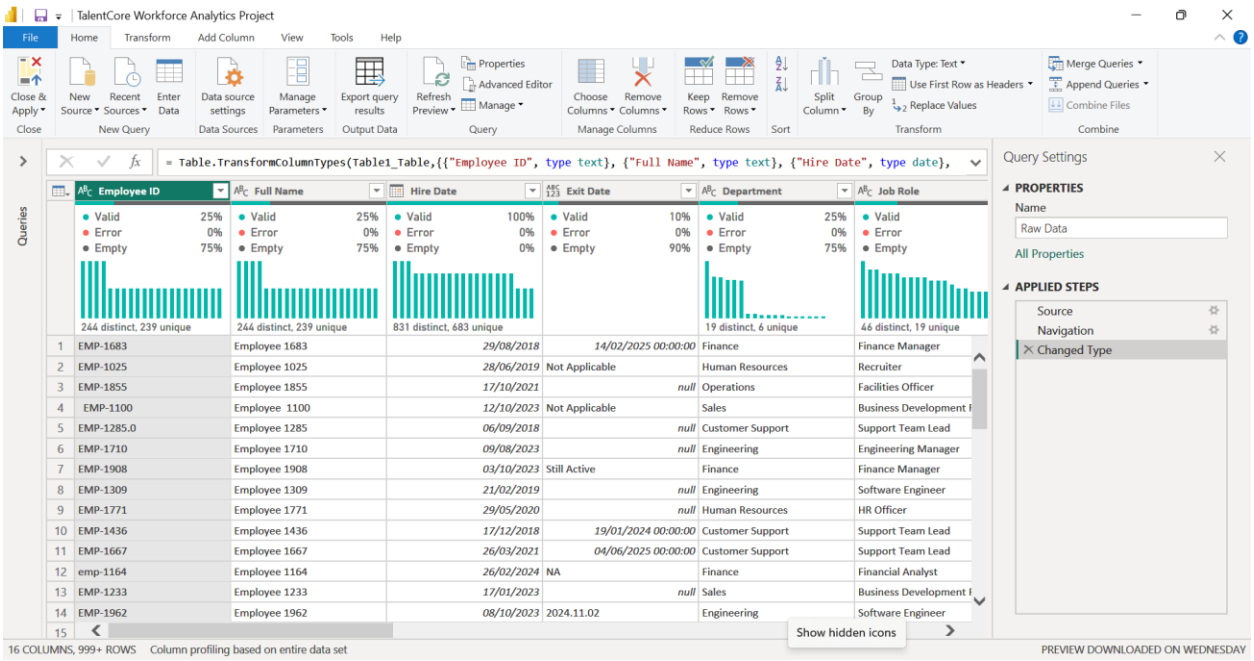


3. The Remarks column only has 16 entries with all other rows empty.
4. Most rows are empty for the exit date field as blank indicates active employees.
5. For the monthly salary field, 8 rows that have all other fields completed are missing salary entries.

6. Row 10 has employee age as 200 which is inaccurate from a realistic standpoint, there's also 2 entries with -5 as the age.
7. All 4 duplicate entries have different hire dates for each entry but similar entry for all other fields.
8. 2 entries have attrition status as left but no exit date recorded for the exit date field which should mean they are still active employees.
9. There were 9 entries that had the exit date earlier than the hire date which implies the exit dates were entered incorrectly. We will still use the entries as the attrition status confirms that they left thus reaffirms validity of the exit dates being provided in reverse order. 9 entries had an Exit Date earlier than the Hire Date and an attrition status of "Left".

Preparation (Data Cleaning and Preparation)

The data preparation phase involved using Power Query to systematically clean the raw data to ensure accuracy and reliability for subsequent analysis.



Preparation Steps Performed on Power Query

1 **Validation:** Checked for spelling mistakes then case inconsistencies were standardized to ensure uniform representation. Trimmed areas where was necessary for cleanliness of the table. Double spaces were removed in some entries on the full name field. Exit Date column was changed to the date data type then all rows that returned an error were replaced with null for accuracy as the entries are still employed.

It also seemed that two rows had their date formats stored as excel date serials so those were converted to dates for accurate presentation.

2 **Null handling:** As a first step, I hid all null rows using the Employee ID column and was left with 247 rows. 6 rows did not have age values , as this cannot be guessed, entries will be documented as 'Unknown'. Row 60 had no exit date but the attrition status was 'Left' with a remark 'Promotion Pending' which strongly contributed to the implication that they are an active employee, because of this their attrition status was changed to 'Active' using a custom column.

5 entries had no performance rating but the entries will be used as is for accuracy of analysis.

8 rows had the monthly salary column finalised to null for accurate representation as they are valid employees, they just don't have their salaries documented.

3 **Handling duplicates:** There were 4 duplicate rows which I deleted to remain with 243 rows of data to work with.

4 **Replacing values:** Several entries across the data set were transformed to allow standardisation across all fields. Entries that had different sentence formats were replaced accurately. Row 77 had their monthly salary entry as -105,900 which was assumed an error so the value was replaced to 105,900

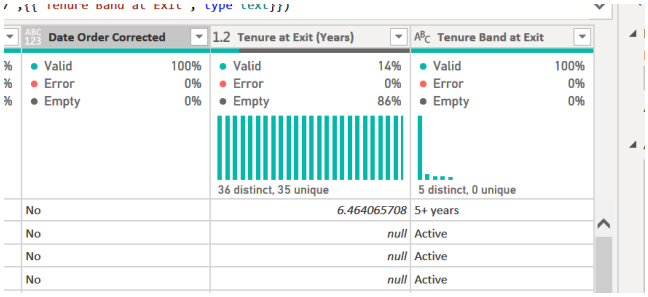
15 rows had their training hours field as N/A, which we replaced to null to avoid coming up with a wrong assumption as using 0 would imply they did no training at all. There were also 3 rows that had negative entries for the column which were all replaced accurately. For the age column, there are 4 entries that have their ages as -5 and 200, given the values are invalid, the inputs were changed to null for accurate representation in the analysis.

6. **Column Deletion:** As there were only 16 entries in the remarks field, the column was deleted as it is also not relevant in the planned analysis. 15 columns remained.

7. **Custom Column:** Based on the full dataset, 37 rows had attrition status as 'Left' despite 2 rows having null in the exit date column. 206 rows showed status as 'Active' which tallies with the null entry in Exit Date for the filter but does not match the total 208 entries when null entries are filtered in the Exit Date column . A custom column was used to accurately update the Attrition Status field using the guideline that a blank Exit Date means an active employee.

I also added 3 custom columns to correct the 9 entries that were assumed to be entered in the wrong order and were swapped, the Date Order Corrected column*(which was removed afterwards)* flag identified these records while the other 2 with the corrected date columns were renamed after the original columns were deleted.

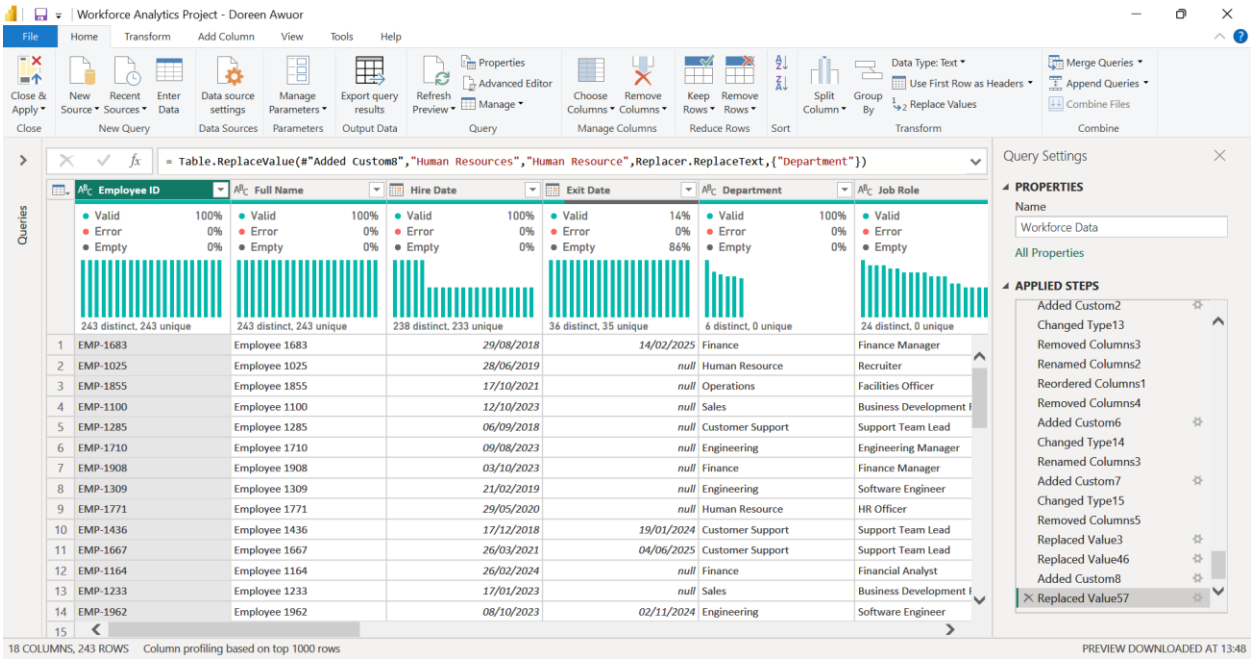
Another custom column for age band was added as this will also be useful in our analysis.



As the null Exit Date rows represents employees who have not left the company, the column is relevant for our analysis despite being 86% empty as records show only 35 employees have left.

The 9% empty cells in the Training Hours column represents the 6 rows that had no entry for training hours completed and 15 rows that originally had N/A as the provided entry. The column is still useful for our analysis of departmental performances thus will be used in the report.

Complete cleaning of my data resulted in 243 rows and 18 columns which will be used in my analysis of the workforce.

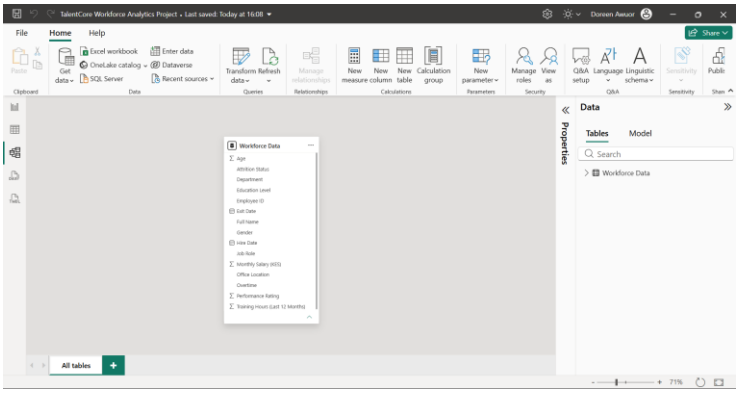


Data Modelling (Data Summarization)

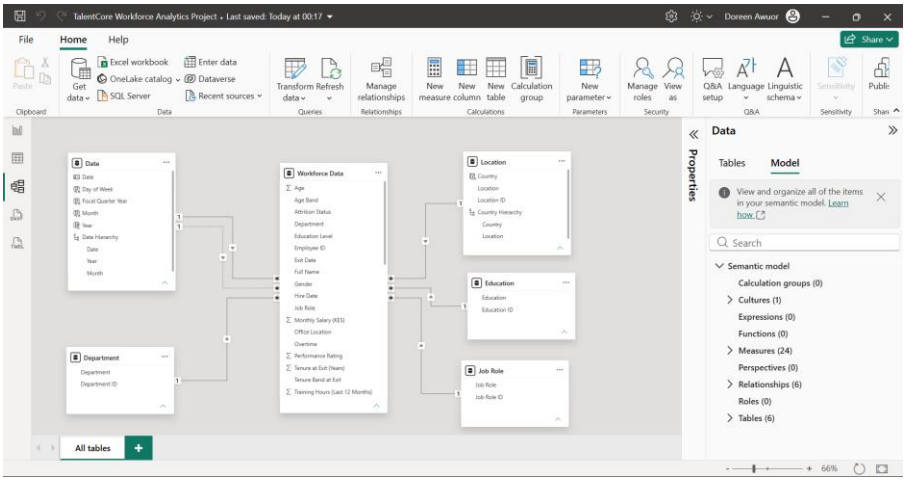
Overall Workforce Insights

KPI	
Current Employees	208
Avg Performance Rating	2.89 out of 5
Avg Monthly Salary	Ksh148,660
Gender Pay Gap	2.69%
Avg Tenure of Leavers (Years)	2.92 Years
Attrition Rate	14.40%

The workforce data provided was a flat schema and was converted to a star schema by normalising the fact table and also creating a date table which was marked as date. Relationships were then created for the fact and dimension tables that would later help in the analysis.



The dimension tables would help in our analysis helping to filter and slice across prepared dashboards.



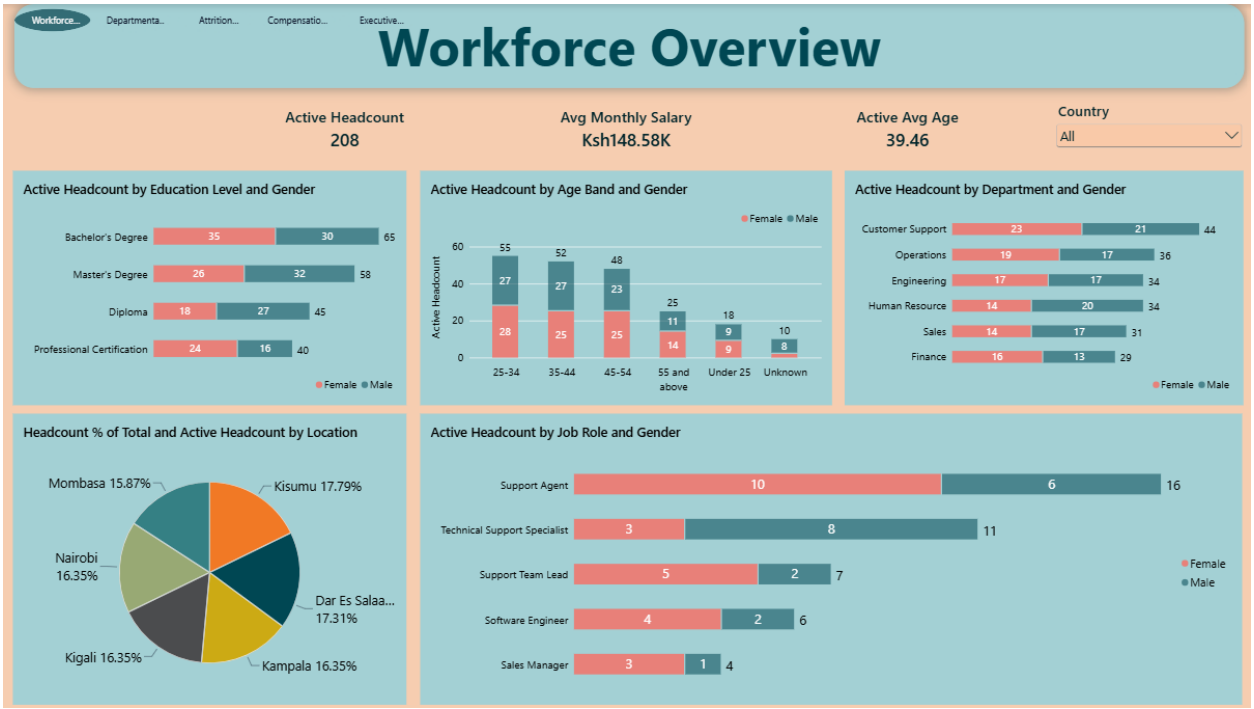
Exits and Avg Tenure of Leavers (Years) depend on the inactive relationship between the Date table and Exit Date thus the inactive relationship in my model. Several DAX measures were also created that would help in the analysis of my objectives.

Measure	Purpose
Total Employees	Unique employees on record, using Employee ID.
Active Headcount	Current employees (Attrition Status = Active).
Headcount % of Total	Each group's share of the active headcount in the current selection.
Active Avg Age	The average age of current employees.
Unknown Age Headcount	Employees whose age was missing or invalid, so excluded records are visible.
Attrition Rate	Shows leavers as a percentage of all employees on record.
Exits	Leavers by their exit date using the inactive relationship between the Date table and Exit Date in the fact table.
Avg Tenure of Leavers (Years)	The average length of service of employees at the point they left.
Early Attrition %	The share of leavers who left within their first year.
Rated Employees	Employees with a valid performance rating.
Unrated Employees	Employees with a missing or invalid rating *(to show how many were excluded)*
Avg Performance Rating	The average performance rating (1 to 5).
Avg Training Hours	The average training hours per employee in the last 12 months.
Training Participation %	Shows the share of employees who completed more than 0 training hours.
Avg Rating (Trained)	The average rating of employees with some training.
Avg Rating (No Training)	The average rating of employees with no training hours.
Overtime Rate	The share of employees who work overtime.
Avg Monthly Salary	The average monthly salary of current employees.
Min Monthly Salary	The lowest monthly salary among current employees.
Max Monthly Salary	The highest monthly salary among current employees.
Total Monthly Payroll	Adds up the monthly salaries of current employees.
Avg Salary Female	The average monthly salary of current female employees.
Avg Salary Male	The average monthly salary of current male employees.
Employees Without Salary	Current employees with no recorded salary.
Gender Pay Gap %	The difference between male and female average salary as a percentage of the male average. A positive value means men earn more.

Evaluation (Data Analysis and Interpretation)

Findings against the Business Questions

1. Current Employee Headcount



The analyzed data shows there is a good gender balance in the workforce, there is only 2 more men than women. The average age of current employees is about 39 years and the workforce's average monthly salary is Ksh. 148,580.

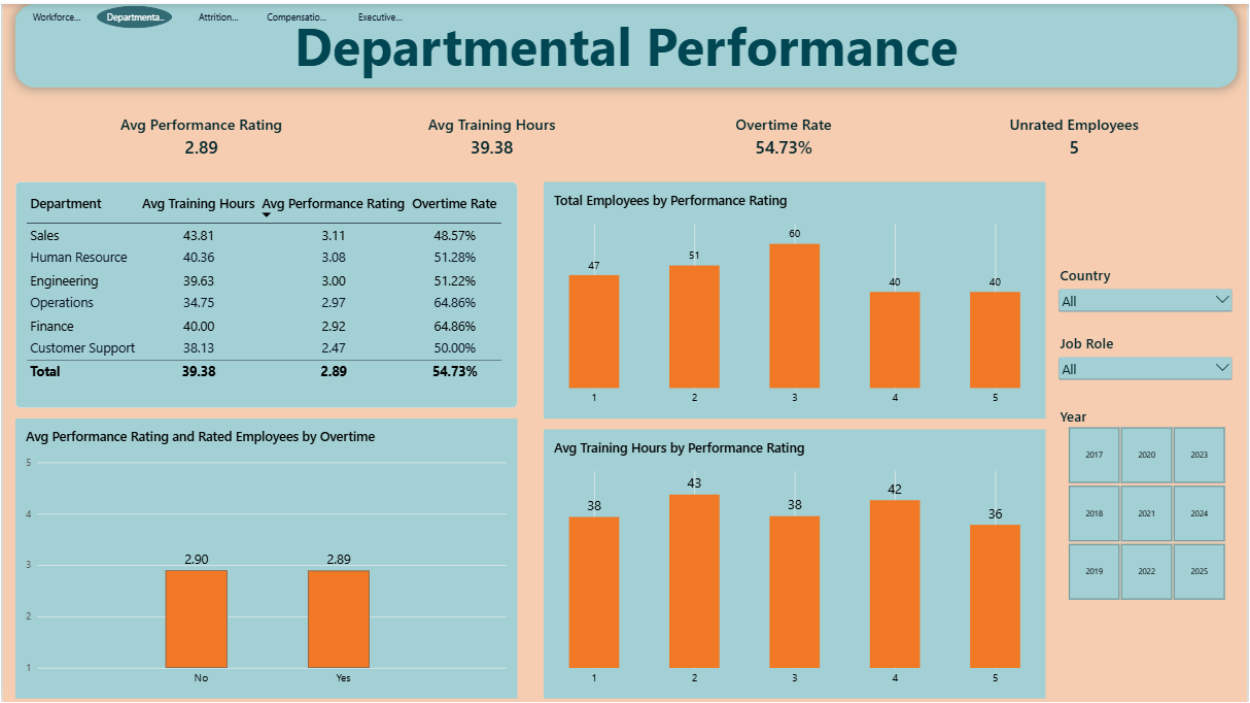
Kenya accounts for 50% of current active employees. Kisumu leads the Kenyan market with 37 employees while Nairobi has 34 employees which is similar to Rwanda and Uganda's active count (16.35%). Tanzania accounts for the remaining 17.30%.

Customer Support leads in the departmental headcount with Finance having the least employees. The age band 25 -34 leads in the headcount (55), the 'Unknown' band represents records that had invalid or missing ages. Using the Top N*(5)* filter, the Support Agent job role has the most staff with majority of them being women.

Bachelor's Degree is the dominant education level with 65 employees, only 40 employees have a Professional Certification. Women lead the headcount for both levels.

Comparing gender by headcount, females account for majority in both the top department, age band, job role and education level. Men on the other hand account for the majority in Human Resource and the Technical Support Specialist job role which has the second-highest headcount. They also account for majority of employees with Diploma and Master's Degree making it an even split in the 4 education levels.

2. Employee Performance Rating and Training Participation



From the data analysed there seem to be no relation between training participation and performance rating. Across departments, employees with less hours of training are also the ones with high overtime rates across all performance ratings. Those with over 40 hours of training seem less likely to do overtime. Most employees fall in the rating 1 – 3 with rating 3 having the highest employee entry (60).

Employees in the ratings 1, 3 and 5 also happen to be the ones with the lowest average training hours.

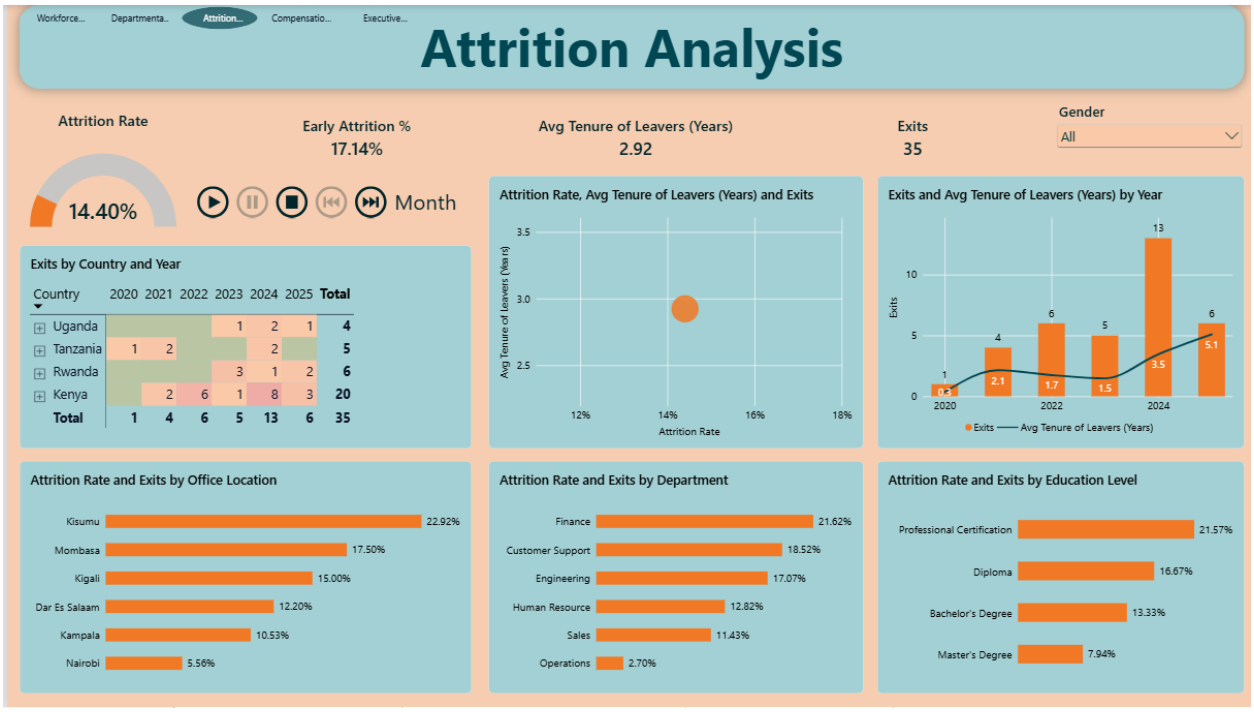
While Sales has the best overall rating with the highest average training hours and lowest overtime rate, it is worth noting that it has the most employees rating between 3 - 5 with over 55% overtime rate. Customer Support has the lowest overall rating but not the lowest average training hours, those with a rating of 1 seem to have the least training hours and the highest overtime (62.50%)

Finance and Operations have an almost similar rating but the notable difference is that for Finance, the lowest rating has the highest overtime rate (71.43%) for all departments with a 1 ranking. Overtime Rate for

Operations seems evenly distributed across all 5 rankings with the lowest overtime rate recorded for the rating 4 (40%). Rankings 1 and 5 have the lowest training hours which supports the earlier conclusion that training participation does not relate directly to performance rating.

Overall, there is minimal impact of overtime on performance rating as results show an almost equal rating whether employees did overtime or not.

3. Average Tenure of Leavers and Employee Attrition Patterns



The company's attrition rate is 14.40% with an average tenure of about 3 years. Kenya has the highest attrition rate(16.13%) of all 4 countries, largely contributed by having 3 office locations. Rwanda follows closely with a 15% attrition rate and 16.67% early attrition. Uganda has the lowest attrition rate (10.53%) and no early attrition. Kenya and Tanzania have a 20% early attrition which is the highest recorded.

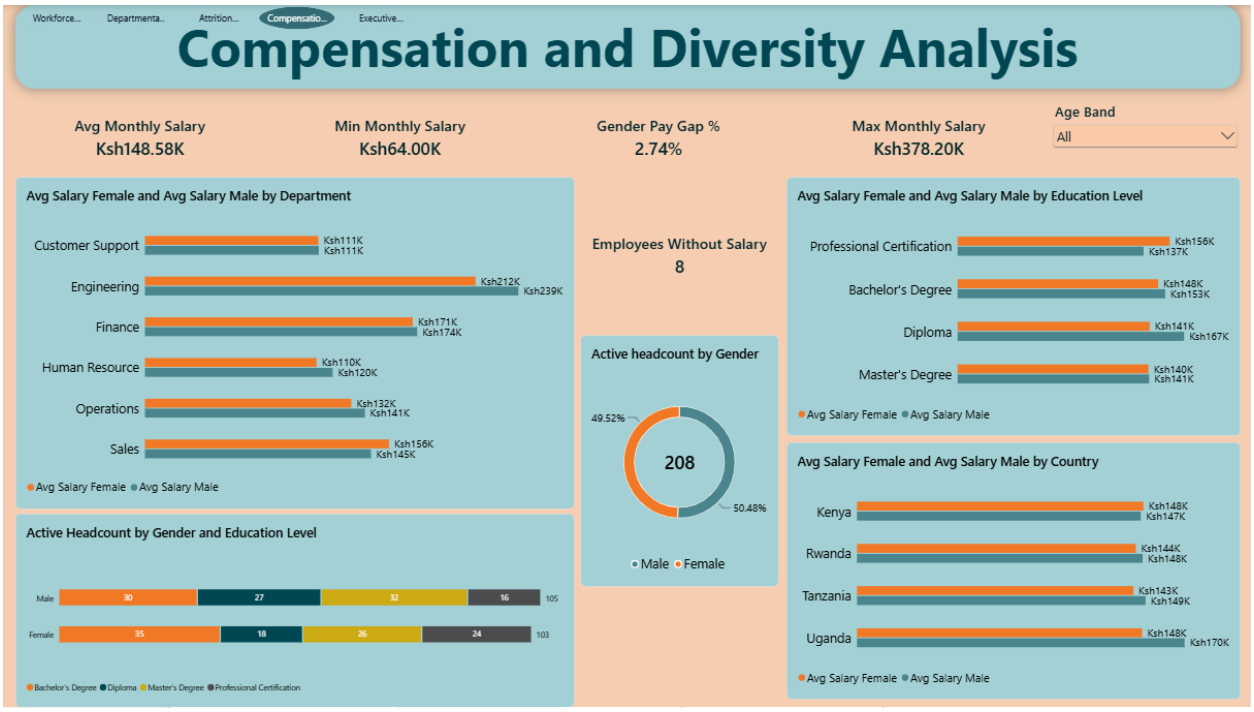
Women are the majority of exits but with an average tenure of about 3 years and an early attrition of 13.04%. While men have less exits, they have a higher early attrition rate (25%) and tenured for about 2 years.

Kisumu has the highest number of exits with an early attrition of 27.27%, while Nairobi has the lowest overall attrition, but a 50% early attrition tied to 2 exits which is the highest of all 6 locations.

Finance department has the highest overall attrition rate with a 25% early attrition. Operations had the lowest attrition rate but 100% early attrition tied to one exit. It is also worth noting that Engineering and Human Resources recorded no early attrition at all.

Diploma holders had the highest early attrition rate despite Professional Certification leading with a 21.57% overall attrition rate. Master's Degree holders had the lowest attrition rate but the second largest early attrition of 20%.

4. Salary Distribution and Diversity Balance in Workforce.



The workforce is quite well balanced with females making 49.52% of current employees. The minimum recorded salary on file is Ksh64,000 and the maximum recorded monthly salary is Ksh378,000. The average monthly salary of current employees with a recorded salary is registered at about Ksh148,580. There are also 8 employees whose salaries are not recorded.

Engineering department has the highest salaries with customer support being the least earners. Male diploma holders are the highest paid on average (Ksh167,000) with females with professional certification come second with an average salary of about Ksh156,000

On a country level, Uganda leads with the highest salary averages for both genders but with a 12.98% gender pay gap, the highest recorded across the four countries. Rwanda and Tanzania's pay gap difference is under

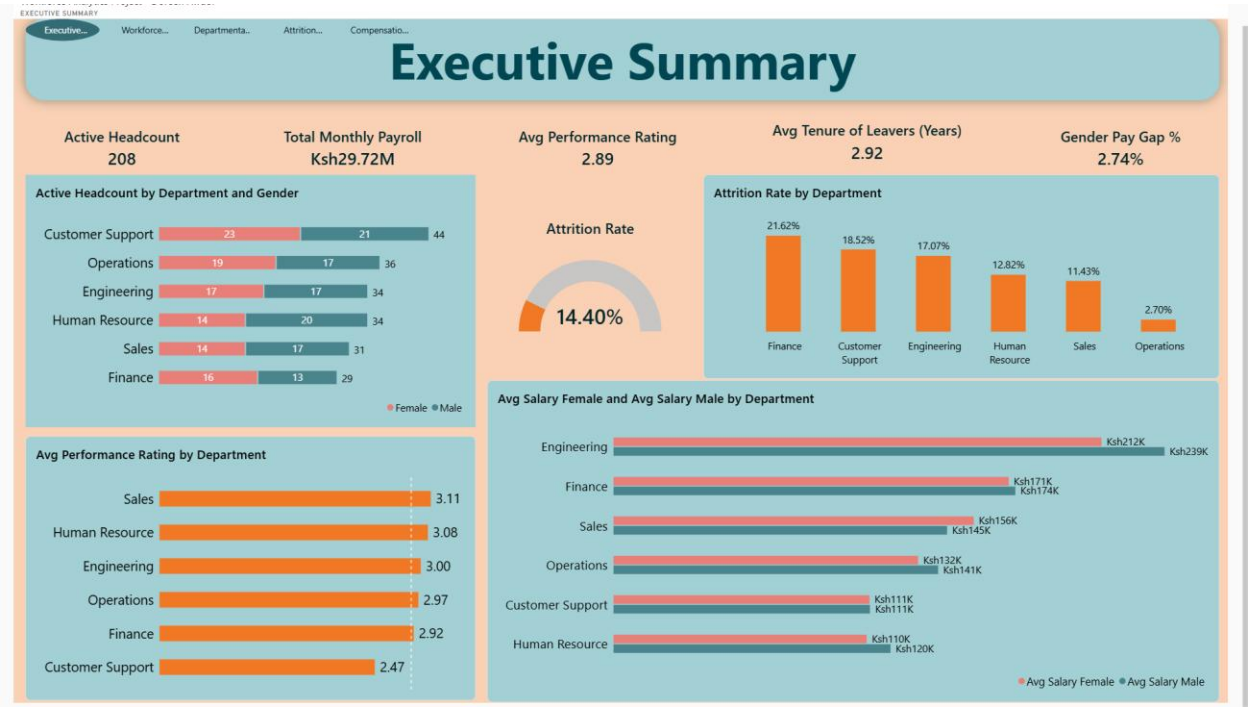
5% while Kenya has a less than 1% gender pay gap in favor of females. In Kenya, the female average salary in Nairobi is the highest for the country with Mombasa having the highest male average salary. Both genders in Kisumu earn on average just about Ksh150,000 making it the most balanced location when it comes to pay.

Males account for most hires by 2 for the current active workforce. Females have the most individuals with Bachelor's Degree and Professional Certification while males record a higher number for Diploma and Master's Degree. Men earn 15.20% more on average than women among Diploma holders, whereas with Professional Certification holders the gap reverses, with women earning 14.10% more than men.

Deployment (Communicating Findings/Dashboard Creation)

This report has five pages *(Workforce Overview, Departmental Performance, Attrition Analysis, Compensation and Diversity Analysis, Executive Summary)* linked by a page navigator. The report was deployed and published on Power BI Service and can be shared as a report with other team members. With BI Service, roles with different rights can be given to assigned relevant individuals.

The dashboard serves as the primary deployment artifact for HR and senior management to understand attrition patterns, monitor compensation equity, evaluate performance, and support strategic talent decisions.



Recommendations

Performance

1. As more training hours show no direct link to performance rating, the training program's content and targets need review. Instead of adding hours focus more on tailored content that is unique to each department and job role. Start with Customer Support department given its lowest rating then Operations next before Finance.
2. Monitor the current workload to see if it is manageable by the current staff or additional help is needed to ensure a better balance that does not result in burn out of employees. Operations and Finance have the highest overtime rate thus are a priority to identify if any changes need to be made within the departments.

Attrition

1. The 2024 spike in exits need to be looked into in detail to better understand where the gap is. Introducing exit interviews or surveys going forward would help further understand why and that would help come up with solutions that drive retention.
2. Target focusing on retention in Kisumu and Mombasa on a location level and Finance and Customer Support on a department level as they have the highest attrition rates in the workforce.
3. First-year onboarding should be improved and strengthened as the findings show 17.14% of leavers left within a year. Improving the onboarding structure and clearly outlining expectations at the onset would help improve early attrition.

Compensation

1. Review pays in the groups with larger gaps by comparing the individual job roles and tenure. From observations, key ones to begin with are Customer Support job roles as the gaps are significant and the average does not accurately represent each role's pay, female Sales Managers earns significantly more on average compared to their male counterparts which may become an issue eventually. Male HR Officers also earn significantly higher compared to the females in the same role. These are just examples of the many scenarios found.

Conclusion

The analysis shows a balanced workforce with moderate overall attrition (14.40%) that is concentrated in specific departments, locations and years. Performance rating average just under 3, with Customer Support having the lowest, and neither training hours nor overtime explain the difference in rating. Pay is broadly equal at company level but gaps in some departments and education levels deserve a closer look. The recommendations given target these areas.